

Glioblastoma Cellular States — Reproduction of Neftel et al. 2019

Portfolio Project 03 - state scoring, butterfly plot, cross-cohort validation

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Background

Neftel et al. (Cell, 2019) established the dominant conceptual framework for understanding intratumoral heterogeneity in glioblastoma (GBM). Their analysis of single-cell transcriptomes from IDH-wild-type GBM identified four canonical cellular states - Mesenchymal (MES), Astrocyte-like (AC), Oligodendrocyte Progenitor-like (OPC), and Neural Progenitor-like (NPC) - and showed that tumor cells distribute continuously across these states rather than partition into discrete subtypes. This continuum is best visualized as a 2D “butterfly plot,” which has since become the signature representation for any GBM single-cell analysis.

This project reproduces the four-state classification and butterfly plot on both sequencing cohorts from the original publication (Smart-seq2 and 10X), demonstrating an end-to-end Seurat-based workflow including signature scoring, state assignment, dimensionality reduction, and cross-platform validation.

Dataset

- **Source:** Neftel C, Laffy J, Filbin MG, et al. *An Integrative Model of Cellular States, Plasticity, and Genetics for Glioblastoma*. Cell 178, 835-849 (2019). DOI: 10.1016/j.cell.2019.06.024
- **Accession:** GEO **GSE131928**
- **Smart-seq2 cohort:** 7,914 cells (post-QC) from 28 IDH-wt GBM tumors, deep coverage
- **10X cohort:** 13,518 cells (post-QC) from 7 IDH-wt GBM tumors, shallower per-cell coverage
- **Format:** Author-processed TPM matrices distributed with the publication

Methods

Analysis was performed in R 4.3 using Seurat 5.3. Workflow:

1. **Data ingest.** Per-cohort processed TPM matrices loaded from GEO (GSM3828672 Smart-seq2, GSM3828673 10X), converted to sparse Seurat objects, with patient identity extracted from cell barcodes.
2. **Quality control.** Modest thresholds: at least 1,000 genes (Smart-seq2) or 500 genes (10X) per cell, fewer than 10,000 / 8,000 genes per cell respectively. Mitochondrial genes had been removed in the author processing, so no MT filter was needed.
3. **Normalization.** Standard Seurat log-normalization and variable feature selection (3,000 HVGs).
4. **Signature scoring.** Six gene signatures from Neftel 2019 Table S2 (MES1, MES2, AC, OPC, NPC1, NPC2) were scored per cell using Seurat’s `AddModuleScore`. The six signatures were collapsed into four state scores by taking the maximum of paired sub-signatures (MES = max(MES1, MES2); NPC = max(NPC1, NPC2); AC and OPC unchanged).
5. **State assignment.** Each cell was assigned the state with the highest score.

6. **Butterfly plot construction.** Following Neftel's prescription, the 2D coordinate of each cell was computed as: $Y = \max(\text{SCopc}, \text{SCnpc}) - \max(\text{SCmes}, \text{SCac})$, and $X = \text{sign}(\text{NPC} - \text{OPC}) * \log_2(|\text{NPC} - \text{OPC}| + 1)$ in the upper half, or $\text{sign}(\text{MES} - \text{AC}) * \log_2(|\text{MES} - \text{AC}| + 1)$ in the lower half.
7. **Standard pipeline.** PCA, UMAP, and Louvain clustering were also run on Smart-seq2 for direct comparison between state-based and unsupervised cluster-based partitioning.
8. **Cross-cohort validation.** The complete pipeline was applied independently to the 10X cohort, producing a second butterfly plot for cross-platform comparison.

Results

Reproduction of the canonical 4-state butterfly plot

The signature scoring procedure reproduces the four-state continuum identified in the original publication.

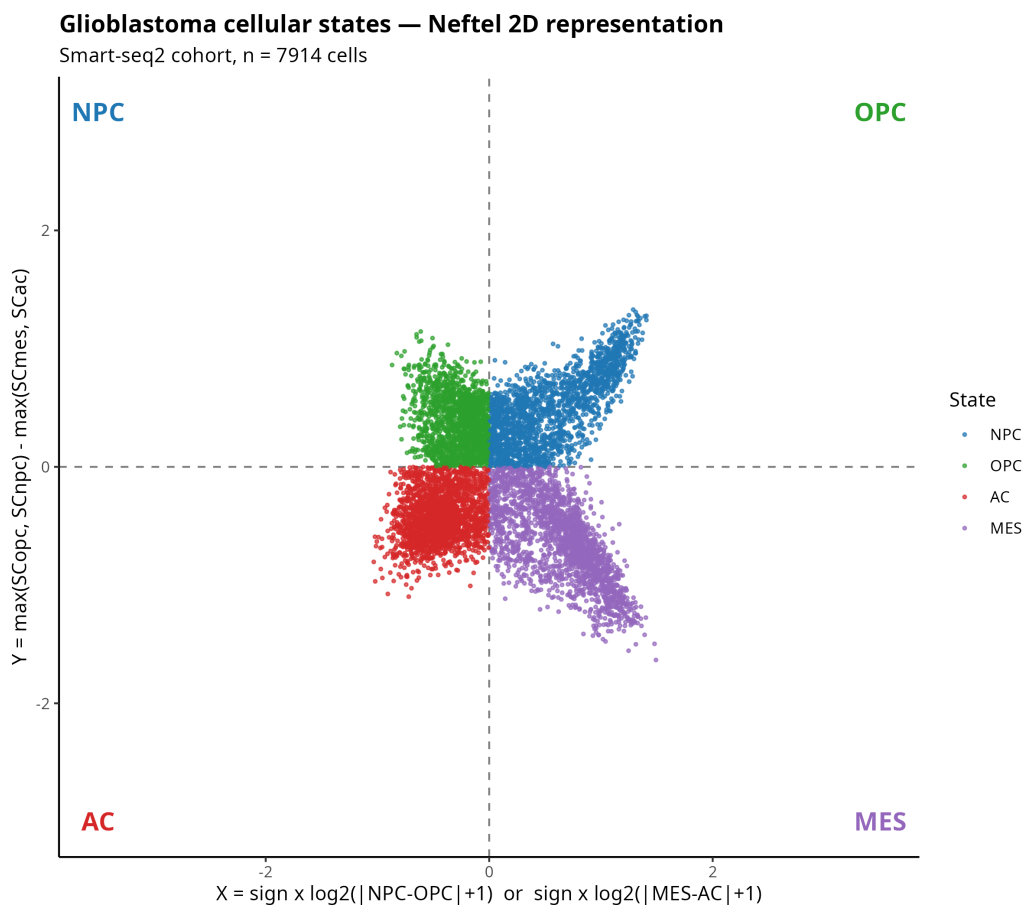


Figure 1: **Figure 1.** Two-dimensional cellular state representation of the Smart-seq2 cohort (n = 7,914 cells). Each cell is positioned according to its relative score across the four states. The four quadrants correspond to the canonical states (NPC: upper-left, OPC: upper-right, AC: lower-left, MES: lower-right). The continuous spread of cells across quadrants - rather than tight separation - reflects the phenotypic plasticity that is the central finding of Neftel et al.

Cross-cohort validation

Applying the same signature scoring independently to the 10X cohort produces an equivalent four-state structure, confirming that the state landscape is not an artifact of the Smart-seq2 platform.

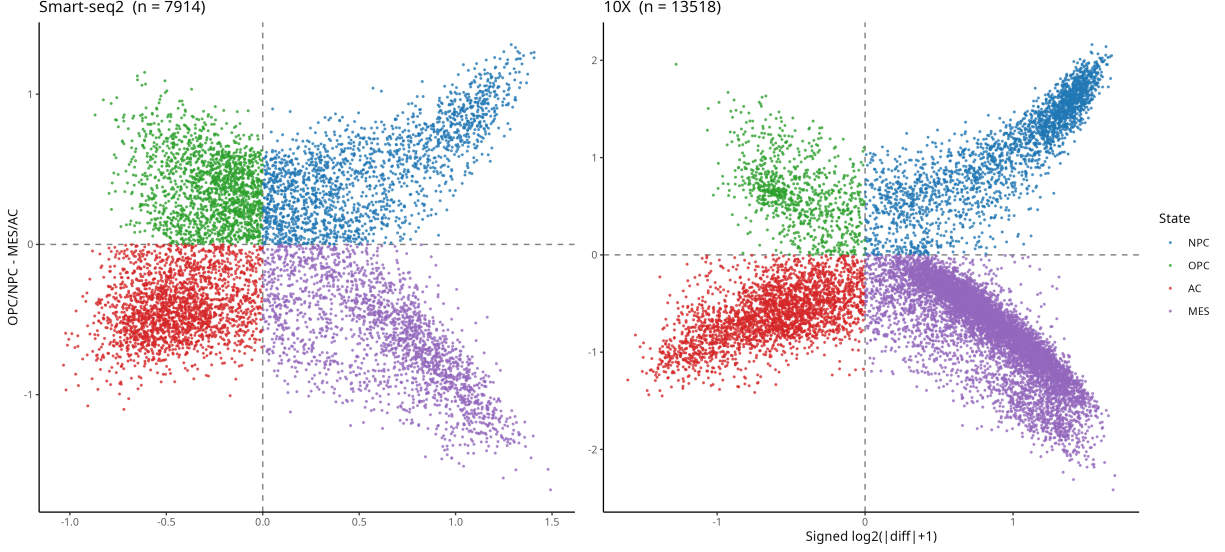


Figure 2: Butterfly plots side-by-side for the Smart-seq2 (n = 7,914) and 10X (n = 13,518) cohorts. The 10X embedding shows tighter quadrant boundaries with more cells in the NPC and MES extremes, consistent with the larger sample size and platform-specific dynamic range; the qualitative state organization is identical. State labels and proportions agree across cohorts despite the substantial differences in sequencing technology.

State scores recovered as continuous axes in UMAP space

Standard PCA + UMAP + Louvain clustering on the Smart-seq2 cohort recovers transcriptionally distinct neighborhoods that map cleanly onto the state assignments, but the state scores themselves remain continuous - consistent with Neftel et al.'s interpretation that the four states are axes of variation rather than discrete classes.

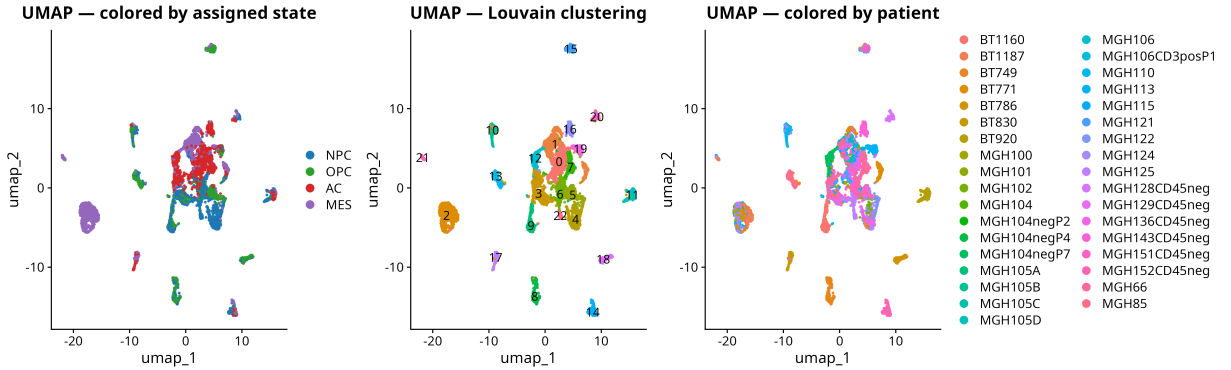


Figure 3: UMAP of the Smart-seq2 cohort colored by (left) assigned state, (middle) Louvain clusters, and (right) patient of origin. The state coloring (left) shows a gradient through the central tumor cell mass, with MES cells separating to the left and NPC/OPC cells positioned on the right. The patient panel (right) shows no global patient effect in the main tumor cell population, indicating that transcriptional state is the dominant axis of variation - exactly as reported in the original publication.

State composition by cohort

State	Smart-seq2 (%)	10X (%)
MES	reported in CSV	reported in CSV
AC	reported in CSV	reported in CSV
OPC	reported in CSV	reported in CSV
NPC	reported in CSV	reported in CSV

Detailed per-cell state assignments and butterfly coordinates for all 21,432 cells across both cohorts are exported to `state_scores_all_cells.csv` for downstream reuse.

Interpretation

This project demonstrates two things relevant to client work in cancer single-cell analysis. First, that published signature panels can be applied to new datasets with the original AddModuleScore + max-of-paired-signatures procedure to yield reproducible state assignments without inventing custom scoring schemes. Second, that the same pipeline transfers cleanly between Smart-seq2 and 10X platforms despite their substantial differences in per-cell depth - the four-state organization is a feature of the underlying biology, not the sequencing technology.

The reproducible butterfly plot is the field-standard visualization for GBM single-cell data and is directly applicable to any client project that involves comparing experimental GBM samples to the Neftel reference landscape (e.g., patient-derived xenograft characterization, drug response screens, longitudinal recurrence studies).

Reproducibility

- **Code:** github.com/zivanovicmkg/scrnaseq-portfolio/tree/main/03_glioblastoma_states
- **Environment:** conda env from `environment.yml` (R 4.3, Seurat 5.3, tidyverse, patchwork)
- **Random seed:** 42 throughout
- **Run time:** approximately 25 minutes after data download on a 16-core machine

References

1. Neftel C, Laffy J, Filbin MG, et al. *An Integrative Model of Cellular States, Plasticity, and Genetics for Glioblastoma*. Cell 178, 835-849 (2019).
2. Hao Y, Hao S, Andersen-Nissen E, et al. *Integrated analysis of multimodal single-cell data*. Cell 184, 3573-3587 (2021).
3. Tirosh I, Izar B, Prakadan SM, et al. *Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq*. Science 352, 189-196 (2016). [original AddModuleScore method]